

```
# Install required libraries
!pip install pandas scikit-learn nltk spacy

# Download spaCy English model
!python -m spacy download en_core_web_sm
```

Requirement already satisfied: python-dateutil<=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas)

Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn)

Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn)

Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.2.1)

Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk)

Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.1)

Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: weasel<0.5.0,>=0.1.0 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: typer<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: pydantic!=1.8,!<1.8.1,<3.0.0,>=1.7.4 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.1.6)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from spacy) (75.1.0)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: language-data>=1.2 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: pydantic-core==2.33.2 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: typing-extensions>=4.12.2 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.12/dist-packages (from spacy)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests)

Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from thinc)

Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.12/dist-packages (from thinc)

Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer)

Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (from typer)

Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from typer)

Requirement already satisfied: smart-open<8.0.0,>=5.2.1 in /usr/local/lib/python3.12/dist-packages (from typer)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2)

Requirement already satisfied: marisa-trie>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from language-data)

Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich)

Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart-open)

Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py)

Collecting en-core-web-sm==3.8.0

Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0.tar.gz 12.8/12.8 MB 80.4 MB/s eta 0:00:00

✓ Download and installation successful

You can now load the package via `spacy.load('en_core_web_sm')`

⚠ Restart to reload dependencies

If you are in a Jupyter or Colab notebook, you may need to restart Python in order to load all the package's dependencies. You can do this by selecting the 'Restart kernel' or 'Restart runtime' option.

```
import pandas as pd
import nltk
import spacy
import string
```

```
from nltk.corpus import stopwords
```

```

from nltk.tokenize import word_tokenize
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

```

```

nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4') # for lemmatizer
nltk.download('punkt_tab') # <-- add this line

```

```

[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
True

```

```

# =====
# Step 1: Load Dataset
# =====
try:
    data = pd.read_csv("Stock Headlines.csv", encoding='utf-8') # Try UTF-8 first
except UnicodeDecodeError:
    data = pd.read_csv("Stock Headlines.csv", encoding='latin1') # If UTF-8 fails, try latin1

# Use only first 100 rows
data = data.head(100)

# Check if required columns exist
if 'Top1' not in data.columns:
    raise KeyError("Column 'Top1' not found in the DataFrame. Please check your CSV file and column name")
if 'Label' not in data.columns:
    raise KeyError("Column 'Label' not found in the DataFrame. Please check your CSV file and column name")

data.head()

```

	Date	Label	Top1	Top2	Top3	Top4	Top5	Top6	Top7	Top8
0	2000-01-03	0	A 'hindrance to operations': extracts from the...	Scorecard	Hughes' instant hit buoys Blues	Jack gets his skates on at ice-cold Alex	Chaos as Maracana builds up for United	Depleted Leicester prevail as Elliott spoils E...	Hungry Spurs sense rich pickings	Gunners so wide of an easy target
1	2000-01-04	0	Scorecard	The best lake scene	Leader: German sleaze	Cheerio, boyo	The main recommendations	Has Cubie killed	Has Cubie killed	Has Cubie killed fees?

```
# =====
# Step 2: Preprocess Text
# =====
def preprocess_text(text):
    text = str(text).lower() # Lowercase
    text = text.translate(str.maketrans('', '', string.punctuation)) # Remove punctuation

    # NLTK Tokenization
    tokens = word_tokenize(text)

    # Remove stopwords
    tokens = [word for word in tokens if word not in set(stopwords.words('english'))]

    # spaCy Lemmatization
    doc = nlp(" ".join(tokens))
    lemmatized_tokens = [token.lemma_ for token in doc]

    return " ".join(lemmatized_tokens)

# Apply preprocessing on the text column
data['clean_text'] = data['Top1'].apply(preprocess_text)
data[['Top1', 'clean_text']].head()
```

5 rows × 27 columns

	Top1	clean_text
0	A 'hindrance to operations': extracts from the...	hindrance operation extract leak report
1	Scorecard	scorecard
2	Coventry caught on counter by Flo	coventry catch counter flo
3	Pilgrim knows how to progress	pilgrim know progress
4	Hitches and Horlocks	hitch horlock

```
# =====
# Step 3: Prepare Features and Labels
# =====
X = data['clean_text']
y = data['Label'] # Assuming binary labels (0/1)

# Convert text to numerical features
vectorizer = CountVectorizer()
X_vect = vectorizer.fit_transform(X)

# Split data into train and test sets
X_train, X_test, y_train, y_test = train_test_split(X_vect, y, test_size=0.2, random_state=42)
```

```
# =====  
# Step 4: Train Logistic Regression  
# =====  
model = LogisticRegression()  
model.fit(X_train, y_train)
```

▼ LogisticRegression ⓘ ?
LogisticRegression()

```
# =====  
# Step 5: Make Predictions  
# =====  
y_pred = model.predict(X_test)  
  
# =====  
# Step 6: Evaluate Model  
# =====  
print("Accuracy:", accuracy_score(y_test, y_pred))  
print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))  
print("Classification Report:\n", classification_report(y_test, y_pred))
```

Accuracy: 0.5
Confusion Matrix:
[[8 2]
 [8 2]]
Classification Report:

	precision	recall	f1-score	support
0	0.50	0.80	0.62	10
1	0.50	0.20	0.29	10
accuracy			0.50	20
macro avg	0.50	0.50	0.45	20
weighted avg	0.50	0.50	0.45	20

Start coding or [generate](#) with AI.